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Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks

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ABSTRACT

This study introduces an innovative approach to detecting electricity theft in smart homes, leveraging a knowledge-based synthetic attack classifier and advanced machine learning techniques. Given the critical challenge of limited labeled data in this domain, our model ingeniously simulates realistic attack scenarios, encompassing a spectrum of electricity theft behaviors. We preprocess five distinct simulated attack classes, extending our model's detection capabilities to unknown anomalies through binary classification. Central to our methodology is the power_base dataset, an aggregation of appliance consumption patterns, which serves as the primary feature for training and evaluating our models. This research uniquely addresses the model's applicability across various residential settings by segmenting anonymized data into three different home categories. Furthermore, the model's efficacy is not only validated through simulated data but also rigorously tested against real building attack data, ensuring its practical relevance and effectiveness. Our selected model, Extreme Gradient Boosting binary-based discriminator algorithm, achieved average AUC scores of 98.74% and 98.69% for real and simulated attacks across various homes; indicating a high level of accuracy in identifying both known and anomalous consumption patterns, positioning our approach as a significant advancement in smart grid security. This study not only contributes to the academic discourse in the field but also offers practical solutions to energy providers and stakeholders in the smart home industry.

INDEX TERMS Electricity theft detection, Machine learning, Synthetic attack data, smart home, Real attack data, Unsupervised learning, and Supervised learning.

I. INTRODUCTION

ELECTRICITY theft is a pervasive issue with economic implications that, necessitate innovative approaches for its detection. Although invisible, it possesses substantial economic value and is an essential resource in modern society. The issue of electricity theft, often referred to as Electricity Theft Attacks (ETAs), represents a pervasive and costly problem worldwide, particularly in developing countries [1], [2]. For instance, in India, it is estimated that more than one-fifth of the total electricity production is lost owing to theft [3]. Moreover, within the United States, these losses are estimated to be approximately \$6 billion annually. [4], [5]. Traditionally, the detection of ETAs relies on human interven-

tion through routine inspections of power meters and cables, user-reported suspicions, or the identification of meter device failures [6]. Nevertheless, these methods are human-centric and struggle to effectively pinpoint instances of electricity theft, primarily because of the inherent complexity of power systems.

Each electrical appliance exhibits a distinctive consumption signature in the time series data, which is aggregated with the consumption patterns of other appliances at the metering point [16]. Consequently, the output of a smart meter encapsulates the collective patterns of electricity consumed by legitimate and potentially malicious loads [17]. This aggregation poses a challenge for the detection of electricity

TABLE 1: Literature Review

Reference	Method	Contributions	Limitations	Our Model's Resolution
Punmiya et al., 2019 [7]	Simple statistical models for synthetic attack loads	Initial frameworks for synthetic load generation and attack simulation.	Limited complexity and realism in attack scenarios.	Novel attack scenarios that simulate realistic electricity theft patterns.
Abraham et al., 2023 [8]	Knowledge-based synthetic attack generation	Understanding the intentions behind various attacks.	Lack of empirical validation with real attack data.	Validation with real attack data for binary classification and anomaly detection.
Dayaratne et al., 2021 [9]; Singh et al., 2017 [10]; Punmiya et al., 2019 [7]	Protection of utility companies from electricity theft	Large-scale protection to prevent revenue loss from power stealing.	Not addressing power theft between individual homes.	Detection and protection against electricity theft at the household level.
Jokar et al., 2015 [5]	Limited Reduction Techniques	Differentiation between benign and attack data in binary classification models.	A gap in knowledge regarding the utilization of appliance consumption patterns for ETD.	Proposes a classifier that uses reduced datasets for efficient and effective theft detection.
Moradzadeh et al., 2020 [11]	Improving residential load disaggregation	Utilized PCA for sustainable energy development.	Did not focus on electricity theft detection.	Utilization of appliance consumption patterns for theft detection.
Singh & Yassine, 2018 [12]	Big data mining for energy consumption forecasting	Advanced forecasting using big data mining techniques.	Lack of application to electricity theft detection.	Data mining of usage patterns for electricity theft detection.
Wang & Yang, 2018 [13]	Iterative load disaggregation	Addressing complexity in appliance-specific patterns.	Focused on disaggregation, not theft detection.	Incorporation of consumption patterns to distinguish theft behavior.
Wilhelm & Kasbauer, 2021 [14]	NILM for human activity recognition	Exploited meter measurements for activity recognition.	Did not apply findings to theft detection.	NILM techniques to identify theft-related irregularities.
Zhang et al., 2014 [15]	Simulation of human living activities	Transformation between activities and power patterns.	Focus was not on theft detection.	Relationship between activities and consumption used for theft detection.

theft through traditional heuristic analysis of meter readings alone [18]. However, with the rapid advancement of machine learning technologies, there is a burgeoning opportunity to harness the power of artificial intelligence to detect electricity theft from aggregated consumption patterns.

This study introduces an innovative Electricity Theft Detection (ETD) framework tailored for smart homes, representing a notable departure from traditional approaches that integrate knowledge-based synthetic attack data. Our approach is particularly well-suited for handling unlabeled datasets, which are commonly addressed using unsupervised learning techniques [19]. Although synthetic data generation is often considered a form of supervised learning owing to the creation of labels through a synthetic approach [20], [21], it is applicable to unsupervised methods when dealing with unlabeled data. However, our research demonstrates the superior performance of our approach compared with conventional unsupervised learning techniques. This finding underscores the primary message conveyed in the present study.

One of the key challenges in deploying machine learning classifiers for ETD is the pervasive issue of data imbalance [22], [23], where the volumes of normal and abnormal samples exhibit substantial disparities. While benign samples are readily accessible through historical data, attack or theft samples are often scarce, and, in some cases, entirely absent for certain customers.

To address this dilemma, we addressed the issue of imbalanced data and zero-day attacks [24] through the creation of a synthetic attack dataset. Leveraging the inherent predictability of theft patterns (Jokar et al., 2015) [5], this

innovative approach significantly elevates the detection rate and empowers the identification of a diverse range of attack types.

A. MOTIVATION

Our motivation for detecting electricity theft in smart homes using appliance consumption patterns is driven by the need to enhance energy security and fairness, reduce financial losses, and ensure reliable and sustainable energy distribution, by leveraging detailed and real-time data generated from smart home appliances to accurately identify irregularities and potential fraudulent activities.

B. LITERATURE REVIEW

Table 1 shows some available literature on appliance consumption, however, there is a significant gap in knowledge regarding the utilization of appliance consumption patterns to train a classifier for electricity theft detection in smart homes. Although there is extensive research on appliance consumption patterns [25], [14], [15] (Singh & Yassine, 2018; Wilhelm & Kasbauer, 2021; Zhang et al., 2014), there is a lack of specific focus on using these patterns to train a classifier for electricity theft detection. Existing literature primarily discusses load disaggregation, human activity recognition, and energy consumption forecasting based on appliance power consumption patterns. However, the direct application of these patterns to train a classifier for electricity theft detection has not been explicitly addressed.

The literature provides insights into the challenges associated with appliance consumption patterns, such as the com-

plexity of identifying appliance-specific consumption patterns and overlapping operation of appliances, which makes event detection difficult [13] (Wang & Yang, 2018).

Additionally, some studies discuss the application of appliance power consumption patterns for simulating human living activities [15] Zhang et al. (2014) and improving residential load disaggregation [11] (Moradzadeh et al., 2020). Furthermore, some studies have emphasized the accuracy of identifying appliance usage patterns using the proposed models [25], [14] (Singh & Yassine, 2018; Wilhelm & Kasbauer, 2021).

However, the specific task of using appliance consumption patterns to train a classifier for electricity theft detection in smart homes remains an underexplored. The references did not directly address the development of a classifier for detecting electricity theft based on appliance consumption patterns. Therefore, there is a clear gap in existing knowledge regarding this specific application.

In summary, although the literature provides valuable insights into appliance consumption patterns and their applications, there is a notable gap in knowledge concerning the direct utilization of these patterns to train a classifier for electricity theft detection in smart homes

Research Gaps

Our analysis reveals several key areas in which existing research falls short, necessitating further investigation and innovation.

Limited Scope of Reduction Techniques: Previous studies, exemplified by the work of Jokar et al. (2015), have primarily focused on basic reduction techniques such as random, mean, and reverse-time-based methods for differentiating between normal and anomalous data in binary classification models. Although foundational, these approaches offer limited complexity in capturing the nuanced dynamics of electricity theft.

Gap in Realism and Validation: There notable deficiency in the simulation of electricity theft patterns that authentically represent the complexities encountered in real-world scenarios.

Challenges in Simulating Sophisticated Theft Scenarios: The literature does not adequately address the challenge of creating advanced and realistic simulations that can accurately emulate the multifaceted nature of electricity theft, which hinders the development of robust detection systems.

Opportunities for Enhanced Anomaly Detection: There is a significant opportunity to improve methods for anomaly detection, particularly by employing real-world attack instances in the validation of models, thus enhancing their predictive accuracy and reliability.

Addressing the Identified Gaps

Our proposed research endeavors to address these identified gaps through several key initiatives:

Incorporation of Complex Theft Patterns: Our study aimed to integrate more intricate and realistic patterns of electricity theft into our model training processes. This approach

is intended to better reflect the diverse and complex scenarios that characterize actual electricity theft incidents.

Validation with Real-World Data: We placed a strong emphasis on validating the effectiveness of our trained models using verified datasets of real-world attack instances. This step is crucial to ensure that our models can accurately predict anomalies and are capable of practical applications in smart home environments.

Advancing Detection Methodologies: Our research introduces more sophisticated methodologies that extend beyond traditional reduction techniques. These advanced methods were designed to provide a more nuanced and comprehensive understanding of the patterns and behaviors indicative of electricity theft.

Contribution to Anomaly Detection in Smart Grid Cybersecurity: By enhancing anomaly detection capabilities through our proposed models and methodologies, our research contributes significantly to the body of knowledge in smart grid cybersecurity. This contribution is particularly relevant in the context of safeguarding smart home environments against electricity theft.

C. CONTRIBUTIONS OF THE PROPOSED MODELS

The development and implementation of our models for electricity theft detection in smart homes has led to several significant contributions to the field.

- **Knowledge-Based Synthetic Attack Classifier:** Our model addresses the challenge of the lack of labeled data in electricity theft detection by implementing a knowledge-based approach. This synthetic attack classifier leverages domain expertise to generate artificial attack scenarios, thereby overcoming the typical data scarcity in this domain.
- **Preprocessing of Simulated Attack Classes:** We meticulously designed five simulated attack classes that reflect various potential theft behaviors. This preprocessing strategy extends the capability of the model to classify unknown patterns, providing a comprehensive approach to anomaly detection.
- **Anomaly Binary Classification:** Utilizing binary classification, our model accurately distinguished between normal and anomalous consumption patterns. This approach is rooted in domain knowledge, ensuring that the classification is both accurate and relevant to the specific context of smart home electricity usage.
- **Aggregation of Appliance Consumption Patterns:** The `power_base` dataset, representing aggregated appliance consumption patterns, serves as the primary feature for training our models. This aggregation captures intricate and varied usage patterns, forming a robust basis for detecting anomalies indicative of electricity theft.
- **Anonymized Data Splits for Different Home Categories:** The dataset was divided into three distinct home categories to evaluate the performance of the models in diverse domestic settings. This categorization not only

enhances the relevance of our models to various real-world scenarios but also underscores our commitment to privacy preservation through data anonymization.

- **Evaluation on Real Building Attack Data:** In addition to the simulated attacks, our models were evaluated using real building attack data. This evaluation provides a practical benchmark for the models, affirming their effectiveness in real-world scenarios and bridging the gap between theoretical simulation and practical applications.

These contributions highlight the innovative aspects of our model, particularly in addressing the challenges of data scarcity, ensuring privacy, and providing a nuanced understanding of electricity consumption patterns in smart homes. Our approach sets a new standard in the field of electricity theft detection, paving the way for more secure and efficient smart home systems.

The remainder of this paper is organized as follows. Section II discusses related work. Section III describes the attack model before proceeding to Section IV to explain our proposal: electricity theft detection with synthetic attack data. Section V describes the experimental setting. Section VI presents the experimental results and performance evaluation. Section VII discusses the model performance comparison, Section VIII provides a discussion and future work, and Section IX concludes the paper.

II. RELATED WORKS

Aldegheishem et al. [26] identified five distinct methods of electricity theft attacks from the perspective of utility companies: (1) meter bypass, (2) meter hacking, (3) direct hooking, (4) tapping, and (5) meter tampering. Additionally, the concept of a False Data Injection Attack (FDIA) for electricity theft [9], [10] has been discussed as a cyber-attack within the realm of cyber-physical systems. Jokar et al. [5] also included FDIA alongside physical attacks that involve bypassing or tampering with meters, all of which manipulate meter readings to reduce electricity consumption [27]. Consequently, the focus of attention has shifted towards electricity theft detection through the analysis of meter readings.

Various machine-learning techniques have been explored for classifying electricity consumption patterns and detecting electricity theft. These methods include Support Vector Machines (SVM) [5], [28], Decision Trees [29], and more recently, Gradient-Boosting [7], [30]. These approaches typically adopt supervised learning principles and, rely on labeled datasets containing both benign and attack data samples. However, obtaining ground truth labels for real-life scenarios is challenging.

An alternative approach to electricity theft detection, particularly when labeled data are unavailable, is anomaly detection. This method encompasses anomaly pattern detection based on hypothesis testing (APD-HT) [32], hierarchical self-organizing maps (SOM) [33], and stacked sparse denoising autoencoders [34]. Anomaly detection, while effective, has the drawback of flagging any abnormal occurrences, includ-

ing instances such as an unusual appliance usage pattern during the nighttime [35].

Our work operates within the context of real-life scenarios where labeled data are scarce. However, we introduced a novel approach by incorporating knowledge of potential attack scenarios and synthetic attack data to train a supervised model using a non-labeled real-world dataset. Furthermore, our work capitalizes on fine-grained time-series data within a smart home environment, a resource that is currently available in today's smart grid landscape.

III. ELECTRICITY THEFT ATTACK

In this section, we consider two categories of electricity theft attack (ETA) model in the context of real-world scenarios, where labeled data are scarce.

First, we overview our synthetic dataset to simulate five attack classes, Figure 1B, to train a supervised model, as mentioned in our previous study [8].

In the second part, we preprocessed our dataset for anomaly detection through binary classification to capture attacks that fell out of the legacy and smart attack thresholds in the previous ETA model. For example, attack parameters lower than 100W or higher than 2000W, as described in Table 5, may not be detected.

A. SYNTHETIC DATASET OVERVIEW

Data Collection

For our study, we utilized the AMPds2 dataset (Almanac of Minutely Power Dataset version 2) [36] as a benchmark, representing two years' worth of home power consumption data. AMPds2 includes minute-by-minute power measurements recorded at the outputs of power distribution boards as shown in Figure 1A. The monitoring points within this dataset are detailed in Table 3. Given the varying configurations of homes, it is possible that certain appliances, such as Clothes Dryers, Wall Ovens, or Dishwashers, may not be present in some households. To address this variability, we simulated three distinct home types by excluding the power consumption of these optional appliances as listed in Table 4.

In configuring each home, we assumed the following:

- **Home A**, some appliances associated with HPE and WOE have peak power consumption, which may allow the peak-hour attacker to steal power more efficiently.
- **Home B** that the existence of a cloth dryer and wall oven may influence the accuracy of the attacker detection.
- **Home C** that the existence of a dishwasher, heat pump, and small appliances may influence the accuracy of attacker detection.

Recognizing that electric power consumption data are inherently time-dependent, we adopted a data segmentation strategy. Specifically, we selected the initial 80% of the data, equivalent to 584 days, for our training dataset. During this phase, we applied our synthetic attack data methodology to enrich the dataset. The remaining 20% of the data, spanning 146 days, was reserved for our test dataset in our benchmark

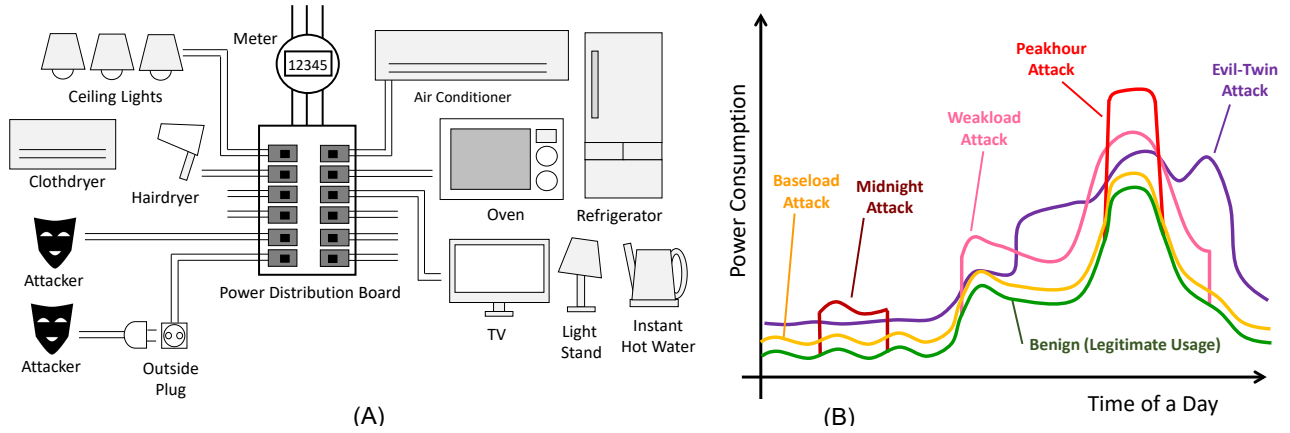


FIGURE 1: A: Power Distribution Board with connected sample appliances. B: Scenarios of electricity theft attacks. Power distribution boards and cables are deployed behind walls and are usually not visible. An attacker uses the pre-deployed cable in a complex building or the outside plug to steal electricity. Depending on the theft pattern, we identify five classes of attacks.

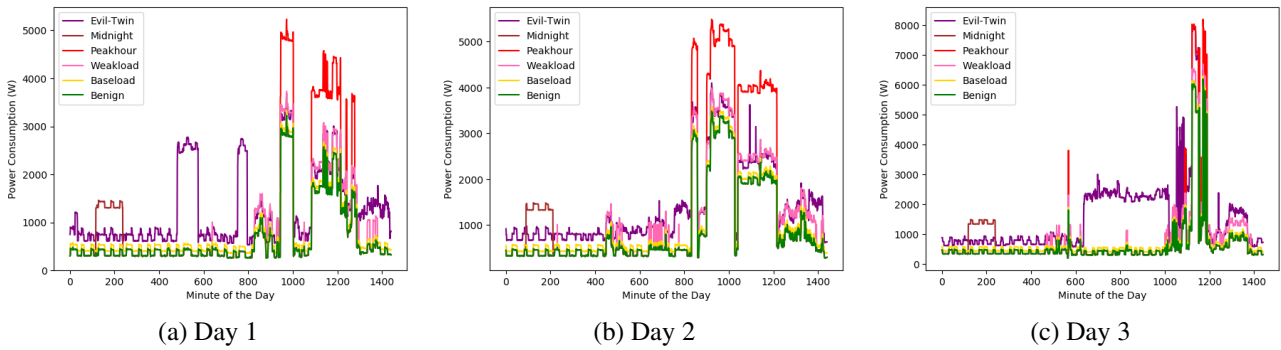


FIGURE 2: Electricity usage of Home A with synthetic attack data on different days. The benign consumption patterns drastically change day by day. Especially, The Weakload and Peakhour attacks are sophisticated – having similar consumption patterns with the legitimate usage although the values are different.

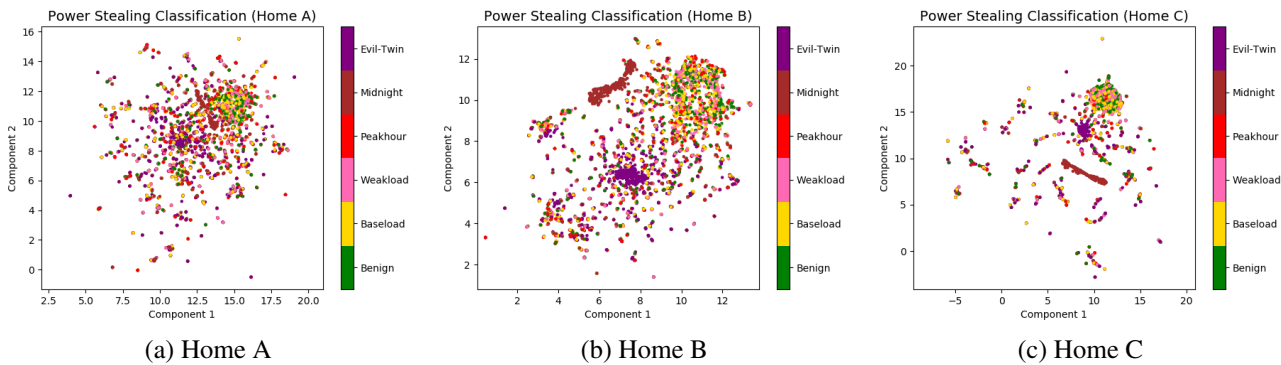


FIGURE 3: 2D projections of attack contained electricity usage with the benign case by uniform manifold approximation and projection (UMAP) [31]. Each plot corresponds to the consumption pattern of a day on different attack scenarios. Depending on the available appliances of the homes, the shapes of the attack classes are very different. There are more overlaps among classes in Home A, whereas we observe fewer overlaps in Home C.

experiment as listed in Table 2. Importantly, the test dataset included simulated attacks generated as part of this study.

It is noteworthy that the test dataset did not cover the entire year. This deliberate choice was made to allocate a larger

volume of data to the training phase, thereby enhancing the robustness of our models.

TABLE 2: The Profile of the dataset generated with synthetic attack data, and corresponding attack impacts (AI)

Home	Category	Benign	Baseload	Weakload	Peakhour	Midnight	Evil-Twin	Averaged AI	AI Ratio
A	Train	584	584	580	490	584	584	1.15 USD	1.77
	Test	146	146	146	140	146	146	1.17 USD	1.72
B	Train	584	584	580	437	584	584	1.08 USD	1.80
	Test	146	146	146	130	146	146	1.10 USD	1.77
C	Train	584	584	576	306	584	584	0.78 USD	1.66
	Test	146	146	146	84	146	146	0.76 USD	1.68

TABLE 3: Appliances and units in AMPDs

Point ID	Description	Point ID	Description
WHE	Whole House Meter	FRE	HVAC/Furnace
B1E	North Bedroom	GRE	Garage
B2E	Master/South BR	HPE	Heat Pump
BME	Basement	HTE	Instant Hot Water Unit
CDE	Clothes Dryer	OFE	Home Office
CWE	Clothes Washer	OUE	Outside Plug
DNE	Dinning Room Plugs	TVE	Entertainment TV/PVR/AMP
DWE	Dishwasher	UTE	Utility Room Plug
EBE	Electronics Workbench	WOE	Wall Oven
EQE	Security/Network	UNE	Unmetered Loads
FGE	Kitchen Fridge		

TABLE 4: Home configurations based on appliance datapoints

Home	Aggregated Appliances	Excluded Appliances
A	B1E, B2E, BME, CWE, DNE, HTE, EBE, EQE, FRE, OFE, OUE, TVE, UTE, CDE, HPE, DWE, FGE, WOE	None (full set)
B	B1E, B2E, BME, CWE, DNE, HTE, EBE, EQE, FRE, OFE, OUE, TVE, UTE, HPE, DWE, FGE	CDE, WOE
C	B1E, B2E, BME, CWE, DNE, HTE, EBE, EQE, FRE, OFE, OUE, UTE, CDE, FGE, WOE	DWE, HPE, TVE

TABLE 5: The Configuration of Synthetic Attack Data For Supervised Learning in the experiment

Attack Class	Configuration
Baseload	Theft = 100W
Weakload	Threshold=500W Theft=500W
Peakhour	Threshold=1500W Theft=2000W
Midnight	Starting from 1 AM - 2 AM (at random) Duration = 120 minutes Theft=1000W
Evil-Twin	One Day was randomly chosen in the dataset.

Data profiles

Table 2 show profiles of the data for our electricity theft attack (ETA) detection study. We have 584 benign records for training each home from the original monitoring data. We simulated and added Baseload, Weakload, Peakhour, Midnight, and Evil-Twin attacks as described in our previous work [8] and parameters threshold for each attack as described in Table

5. In some conditions, if the aggregated base power consumption of the home does not reach the threshold, Weakload and Peakhour attacks (smart attacks) are not triggered.

Figure 3 illustrates the data samples using the Uniform Manifold Approximation and Projection (UMAP) technique for homes A, B, and C [31]. Each plot in this figure corresponds to the data from a single day. Notably, the different homes exhibited distinct data characteristics. For example, in Home A, benign data are dispersed and overlap with various attack samples. Conversely, Home C displayed more discernible clusters which may facilitate classification.

Attack Impact

We define attack impact (AI) as a score to measure electricity theft. This attack affects the amount of energy stolen in the bill.

Let w_i be the weighted price of the power of x_i on the day at each index i . Let us denote the vector of w_i by $\mathbf{w}$.

The attack impact of electricity theft of the day is :

$$I = \mathbf{w} \cdot \mathbf{x}_A = \sum_i w_i x_{Ai} \quad (1)$$

Here, x_{Ai} represents the power stolen at index i of $\mathbf{x}_A$.

To calculate AI on our dataset, we assumed 0.20 USD/kWh constantly for the unit price. The average daily bill mounted by the attacker was approximately 1 USD on average (see Table 2). It is approximately 30 USD monthly and 360 USD per year. If the base power consumption of the home is larger, such as in the case of an office, shop, restaurant, or factory, the attacker will be able to steal more power.

B. DATASET PREPROCESSING FOR BINARY CLASSIFICATION

The electricity theft attack detection dataset (ETA-DD) consists of:

- 1) two training sub-datasets and
- 2) two testing sub-datasets for homes A, B & C

This ETA-DD is assumed for binary classification problems (*Benign* or *Attack*). This is because it is intended for applying unsupervised learning for attack detection and evaluating detection accuracies with real building data. The real building power consumption cannot be easily labeled for the predefined attack cases (as mentioned in subsection A above). These are the reasons why this study focuses on attack detection, rather than classification. Even though it is not intended for attack classification, the evaluation scope has drastically

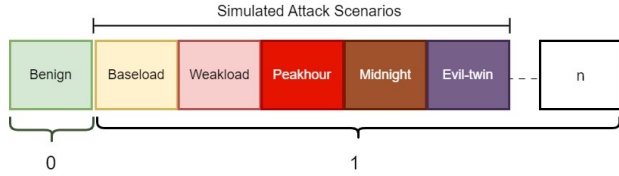


FIGURE 4: Multiclass attack scenarios with unspecified attacks n , preprocessed to binary attack class for anomaly detection.

widened. This paper expands the limitations of the model to detect attacks that are not classified hence we preprocessed our dataset to accommodate the 5 simulated attack scenarios to include unknown or new attacks, n , as 1 and the benign or non-attack data as 0 as shown in Figure 4.

If we denote the original class labels by C where $C \in \{\text{Benign, Baseload, Weakload, Peakhour, Midnight, Evil-twin, } \dots, n\}$, with their attack parameters as shown in Table 5, the binary classification function $f(C)$ can be defined as

$$f(C) = \begin{cases} 0 & \text{if } C = \text{Benign} \\ 1 & \text{otherwise} \end{cases} \quad (2)$$

Here, label 0 corresponds to the normal (benign) class, and label 1 corresponds to any kind of attack scenario. This binary labeling strategy is a common approach in anomaly or intrusion detection systems [37] where the focus is on differentiating between normal and abnormal behaviors, regardless of the specific type of abnormal activity.

Our real test attack data which have the same features (aggregated power_base consumption patterns), further solidify the evaluation of our model's performance.

We extended our research by transitioning from multiclass [8] to binary classification, incorporating data from three distinct homes (Home A, Home B, and Home C as depicted in Table 4).

C. ATTACK SCENARIO

Each house has an electric meter for accounting purposes. The household must pay a bill based on the accumulated power measured by the meter. An attacker may physically access the power distribution board under the electric meter and steal power from the house to make the household pay the bill on their behalf. An attacker may also steal power from an outlet if it is physically accessible as depicted in Figure 1.

This type of scenario may occur in apartments, congested complex buildings, and/or office rooms, where power line cables are not easily traceable. For example, some houses may be leased with pre-deployed lines for electricity theft.

1) Knowledge-Based Attack Simulation Framework

The framework proposed in Figure 5, employs a knowledge-based approach to generate synthetic attack scenarios on power consumption data, encapsulating domain expertise within its operational paradigm. The core of this methodology

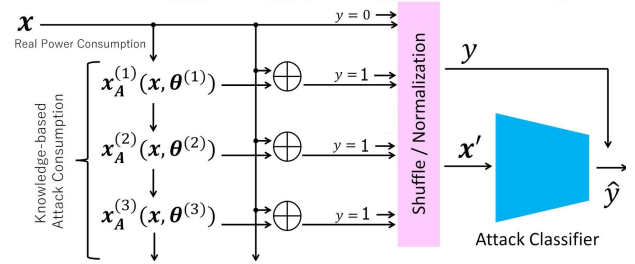


FIGURE 5: ETD framework for processing real power consumption data to detect anomalies that indicate cyberattacks by transforming the data, labeling, preparing for analysis, and finally classifying it using a trained attack classifier.

is the utilization of the actual power consumption profiles, denoted as x , as the foundational dataset from which attack patterns are derived.

2) Attack Data Generation

Distinct attack scenarios are simulated through a series of transformations applied to real consumption data, parameterized by θ . These transformations— $x_A(x, \theta^{(1)})$, $x_A(x, \theta^{(2)})$, and $x_A(x, \theta^{(3)})$ —are crafted based on expert insights into the modus operandi of various attack vectors, with each θ iteration representing a unique attack typology.

3) Data Labeling and Preprocessing

Further cementing its knowledge-driven architecture, the framework classifies consumption data into normal ($y = 0$) and anomalous ($y = 1$) states, employing pre-established criteria that delineate normalcy from theft-related anomalies. Before classification, data undergo a shuffling and normalization process, which is critical for eliminating potential classifier bias attributable to sequential order or feature scale disparities.

4) Attack Classification

The culmination of the framework is the attack classifier, a predictive model trained on a rich historical corpus comprising known instances of consumption patterns, both benign and malignant. This classifier is not merely a data-driven algorithm but a knowledge-infused system tuned to recognize and react to the subtle intricacies of electricity theft within smart grid environments.

The framework's reliance on domain-specific knowledge for the generation and processing of data points designates it as knowledge-based. This is exemplified by the methodical application of expert understanding to the identification of theft signatures, which is paramount for effective discrimination between legitimate and fraudulent electricity usage patterns.

Our knowledge-based framework sets a new benchmark for electricity theft detection systems, marrying the depth of domain knowledge with the rigor of machine learning classification. This synergy promises a robust and discerning

methodology, poised to advance the state-of-the-art in smart grid security.

IV. ELECTRICITY THEFT DETECTION WITH SYNTHETIC ATTACK DATA

In our quest to effectively detect electricity theft attacks, we adopt a comprehensive approach that closely monitors power consumption patterns. We emphasize the importance of granularity in both the time and power domains, particularly at the power aggregation point as shown in Tables 4 and 5, and Figure 6. In this study, we delve into the realm of synthetic and real attack data, leveraging their combined potential to validate machine learning applications. Our primary objective is to accurately identify genuine anomalies originating from legitimate electricity consumption patterns, as shown in the framework of synthetic attack learning for ETD in Figure 5.

Let us consider $\mathbf{x}$, a vector of power consumption. This vector contains the power consumption of each timeslot in time order. For example, $\mathbf{x}$ may represent the power consumption of a certain day, and the i -th element x_i corresponds to the power usage at i -th minute from the beginning of the day. In this case, $\mathbf{x}$ had 1440 elements, that is, $60 \times 24 = 1440$.

In supervised learning, we assume that each $\mathbf{x}$ has a corresponding label y to train a classification model for the power consumption patterns. In our case, we consider unsupervised learning; therefore, we can assume that label $y = 0$ as a benign case, and all other attacks such as the Baseload attack, weakload attack, and, other attacks are all labeled $y = 1$ as an attack case. In real practical scenario, we will only obtain a collection of $\mathbf{x}$ from a house as a result of long-term monitoring, and we will not obtain real attack-enabled cases as anomalies. However, many power-stealing cases can be simulated by arithmetically adding stolen power as power consumption.

Let $\mathbf{x}_A$ be a vector of the stolen power of an attacker. As we assume that the attacker changes the stealing power based on the consumption of the house, $\mathbf{x}_A$ is a function of $\mathbf{x}$ and attacking parameters θ : e.g., $\mathbf{x}_A(\mathbf{x}, \theta)$.

In the case of an attack, that is, $y = 1$, we consider injecting an attack randomly from attack type z . Then, the stolen attack vector is: $\mathbf{x}_A^{(z)}(\mathbf{x}, \theta^{(z)})$.

Finally, we obtain the binary classification (benign and attack) dataset as follows:-

$$(\mathbf{x}', y) = \begin{cases} (\mathbf{x}, 0) & y = 0 \\ (\mathbf{x} + \mathbf{x}_A^{(z)}(\mathbf{x}, \theta^{(z)}), 1) & y = 1 \end{cases} \quad (3)$$

Machine learning models can be applied to the collection of $(\mathbf{x}', y)$ for binary classification and outlier detection.

The proposed framework in Figure 5 benefits from recognizing both specific attacks and unclassified anomalies, integrating the strengths of both approaches for a robust security posture.

Our research is propelled by the crucial need to equip home operators with the ability to effectively detect and categorize cyber threats. Hence the following benefits are manifold:

- **Rapid response:** Our system enables operators to recognize threats immediately by employing binary classification techniques, prompting a quick response to cyber incidents.
- **In-depth analysis and prevention:** Through multi-class classification, we provide comprehensive insights into the nature of attacks, which is essential for effectively strategizing prevention and allocating defensive resources.
- **Strategic planning & resilience:** Our approach to specific attack classifications underpins the development of customized incident response strategies, thereby fortifying system resilience.
- **Compliance and confidence:** Our method ensures compliance with cybersecurity regulations, thereby reinforcing customer trust in the protective measures.
- **Operational integrity:** The efficiency of our detection and classification system is paramount for sustaining operational continuity and mitigating the impacts of cyber threats.

Figure 6 depicts the architecture of the proposed ETD mode to consolidate appliance consumption data from smart homes, ensuring privacy through anonymization and consistency through normalization. Simulated theft scenarios enhance the dataset, with each instance labeled as normal or fraudulent as described in Section III.

A training set derived from this data trains algorithms to detect consumption patterns, whereas a testing set comprising simulated and real data evaluates accuracy. The model employs both traditional machine learning and neural network classifiers, benchmarked against metrics such as accuracy, area under curve (AUC), and F1-score. Post-validation, the model was deployed, with ongoing retraining to refine its detection capabilities.

Data anonymization

Data from various appliances are combined to create a comprehensive view of a home's power usage, with measures to protect user privacy.

The main feature of our dataset is the aggregation of the power consumption of each appliance in each home. The values are only numerical readings without any direct personal identifiers.

The aggregated baseline power consumption for each home (Table 4), $P_{\text{total base}}$ for n appliances is given by the sum of individual baseline power consumptions $P_{\text{base},i}$:

$$P_{\text{total base}} = \sum_{i=1}^n P_{\text{base},i} \quad (4)$$

where $P_{\text{base},i}$ is the baseline power consumption of the i -th appliance.

The total energy consumption E_{total} , considering the duration of usage D_i for each appliance, is calculated as:

$$E_{\text{total}} = \sum_{i=1}^n (P_{\text{base},i} \times D_i) \quad (5)$$

where D_i is the duration of usage for the i -th appliance in hours, and $P_{base,i}$ is as defined earlier.

Normalization

In our ETD for smart homes, we deployed both StandardScaler and MinMax to normalize the feature vectors before synthetic binary discriminators (SYN) and legacy unsupervised models (LUM) experiments respectively.

- For SYN-supervised models, we used StandardScaler to normalize the feature vectors by removing the mean and scaling to unit variance which helped improve the performance and stability of our models. The scaler is first fitted on the training data:

$$\mu_j = \frac{1}{n} \sum_{i=1}^n X_{train_{ij}} \quad (6)$$

$$\sigma_j = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_{train_{ij}} - \mu_j)^2} \quad (7)$$

where μ_j and σ_j are the mean and standard deviation for each feature j , and n is the number of training samples. The training data are then transformed using these parameters:

$$X_{train_scaled_{ij}} = \frac{(X_{train_{ij}} - \mu_j)}{\sigma_j} \quad (8)$$

The same transformation is applied to the test data:

$$X_{test_scaled_{ij}} = \frac{(X_{test_{ij}} - \mu_j)}{\sigma_j} \quad (9)$$

This ensures that both training and test data are on the same scale.

- MinMax Normalization for Anomaly Detection
For our LUM, normalization through MinMax ensures that the input features contribute equally to model training, enhancing the learning process. The MinMax Scaler linearly transforms each feature to a common scale, typically between 0 and 1. The transformation is defined as:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (10)$$

where X_{min} and X_{max} are the minimum and maximum values of the feature in the training dataset, respectively, and X represents the original feature value.

A. DATA PREPARATION

Feature Vector (X) and Label (Y)

Each file starts from a header row, followed by consumption records. Each consumption record is organized as shown in Figure 7.

X : features consisting of 1380 elements, each corresponding to the power consumption of the minute of the day. However, the 1380 min was 23 h. This is because we applied

Data Augmentation, as described in the Data Augmentation subsection below.

Y : label that identifies the meaning of the X . *Benign* is defined as 0. *Attack* is defined as 1.

Data Augmentation

Let X be the original feature, Table 2, vector of length $n = 1380$, representing the power consumption readings for every minute of 23 h as depicted in Figure 7:-

$$X = [x_1, x_2, \dots, x_n] \quad (11)$$

Let k be the time offset for circular shifting, where $k \in \{0, 1, \dots, 59\}$.

The augmented feature vector X' after applying a circular shift of k positions is defined as:

$$X' = [x_{(i-k) \bmod n}, x_{(i-k+1) \bmod n}, \dots, x_{(i-1) \bmod n}, x_i, \dots, x_{(i-k-1) \bmod n}] \quad (12)$$

where $i = 1, 2, \dots, n$ and x_i is the power consumption at the i -th minute. The modulo operation $\bmod$ ensures that the index wraps around the vector when the shift exceeds the start of the vector.

The label Y remained unchanged for all augmented records. If the original record is labeled as benign ($Y = 0$) or attacked ($Y = 1$), then all augmented versions of the record retain this label:

$$Y' = Y \quad (13)$$

For each record in the dataset, this augmentation process was repeated 60 times as shown in Figure 8, corresponding to each possible offset k , thereby effectively increasing the number of benign records.

The length of the augmented feature vector X' is 1380 instead of the full 1440 which represents the total number of minutes in 24 hours. The reason for this is twofold:-

- 1) A circular shift [38] by an offset k ranging from 0 to 59 min implies that we need to have a buffer at the end of the vector to accommodate the maximum possible shift without wrapping the data of the next day.
- 2) By limiting the feature vector to 1380 min, we ensured that for the largest shift of 59 min, the augmented data still represents a continuous 23-h window from the same day. This is crucial for maintaining the integrity of the daily patterns in power consumption without mixing data from two different days.

The label Y , which represents whether the original 1440-minute vector corresponds to a benign or attack pattern, remains associated with the corresponding 1380-minute augmented vector X' . This ensures that the model learns to detect anomalies based on the most representative and complete daily consumption patterns possible within the constraints of the data augmentation process as listed in Table 6.

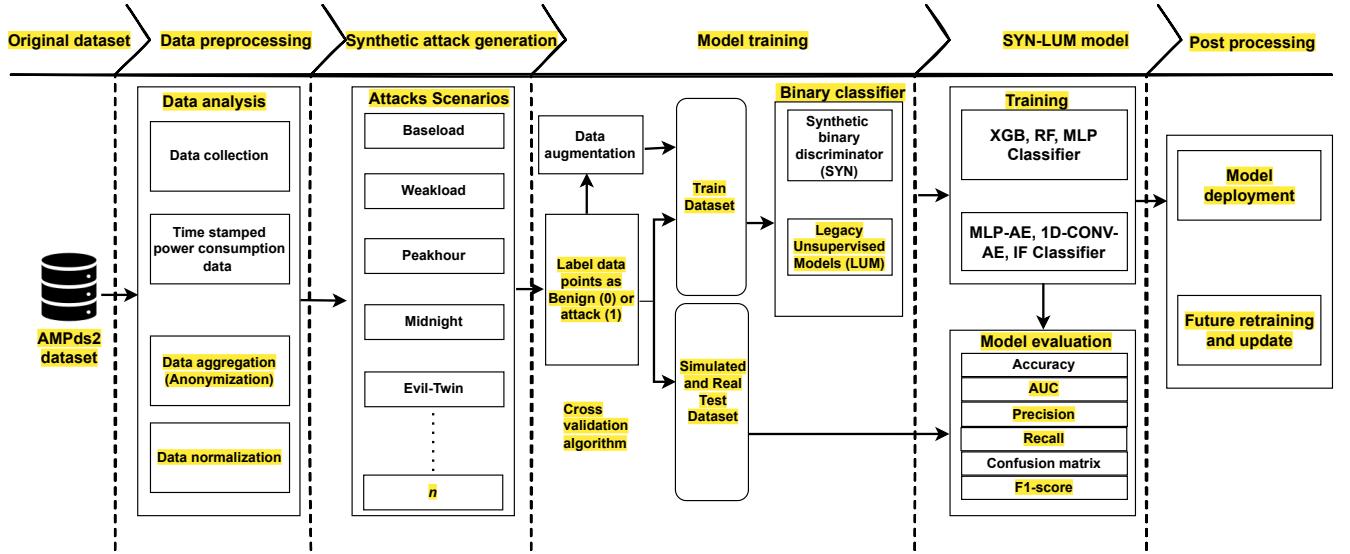


FIGURE 6: Flow of ETD model

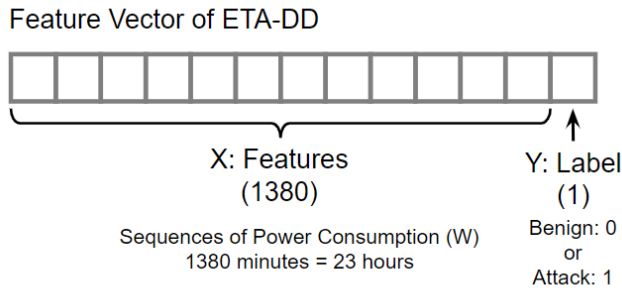


FIGURE 7: Features vector and labels

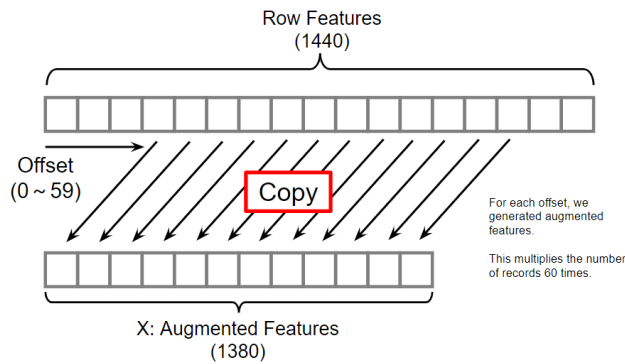


FIGURE 8: Framework of data augmentation

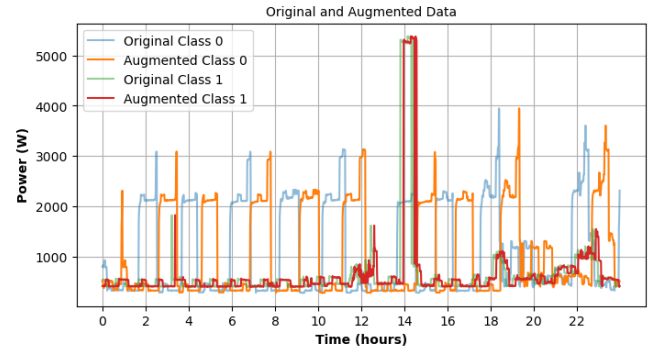


FIGURE 9: Electricity power consumption pattern of Home A for original class and augmented data class for a day

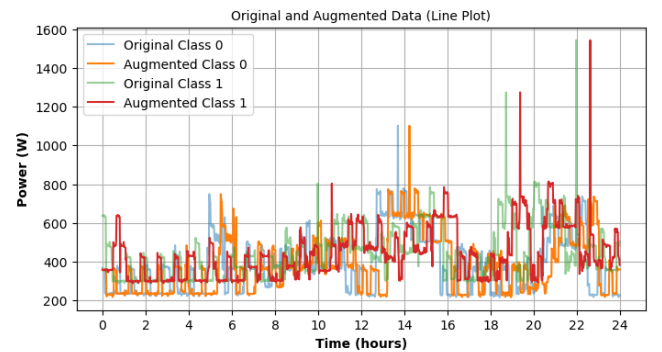


FIGURE 10: Electricity power consumption pattern of all homes for original class and augmented data class for a day

To illustrate the effectiveness of this augmentation technique, Figures 9 and 10 show augmented power consumption patterns for a single Home A and multiple homes (Homes A, B, and C), respectively.

In our daily lives, the operation of home appliances may

be shifted by approximately one hour. Based on this idea, we have shifted the original data over the time axis up to 60 min and extracted them as benign records.

Table 6 introduces the additional data details and param-

eters for ETA detection. We expanded the number of benign records to 35,040 benign training records, each home, sourced from the monitoring data. The table reveals that there were "0" instances of unsupervised training data across all homes. The LUM anomaly detectors were trained solely on benign samples and then subjected to testing on datasets containing both benign and simulated attack samples while synthetic supervised detectors were trained with both benign and attack samples. Their performance was further evaluated using real attack samples.

In general, if the AI of energy theft is large, it can be easily detected with higher accuracy. If the AI of the energy theft is small, it can be negligible, that is it does not need to be detected. Therefore, we selected well-balanced power consumption for evaluation from real building facilities.

TABLE 6: Distribution of simulated and real binary dataset

Dataset	Home	Benign samples	Attack samples
SYN Train	A	35040	33818
	B	35040	33242
	C	35040	31533
UN Train	A	35040	0
	B	35040	0
	C	35040	0
Sim Test	A	146	144
	B	146	142
	C	146	133
Real Test	A	8760	8760
	B	8760	8760
	C	8760	8760

Note: SYN = Synthetic, UN = Unsupervised

UTokyo Data - Real Attack Data

To test the performance of ETD in both synthetic and unsupervised approaches, we consider including the consumption pattern of real rooms of the University of Tokyo as attack data for the model tests. We measured the power consumption at the power distribution boards of the I-REF building (6th-floor building) from 2012 with a sampling frequency of 1 min. We selected the daily consumption patterns in which the attack impact (AI) corresponded to the attack on the simulation test dataset.

B. APPLICATION OF ETD BASED BINARY DISCRIMINATOR

Suppose X represents the input features, (Figure 11), from the synthetic train dataset, and Y is the output prediction for binary classification,

$$f(X) = Y \quad (14)$$

where f represents the learning function of the Binary Discriminator.

During the testing phase, the trained model, f is evaluated using two different datasets:

For the Simulated Test Dataset:

$$f(X_{sim}) = Y_{sim}^t \quad (15)$$

where X_{sim} are the input features and Y_{sim}^t is the output predicted by the model.

For the Real Test Dataset:

$$f(X_{real}) = Y_{real}^t \quad (16)$$

where X_{real} is the input feature and Y_{real}^t is the output predicted by the model.

The performance of the model was assessed based on the accuracy of the predictions Y_{sim}^t and Y_{real}^t in comparison to the true labels.

Proposed models Characteristics overview

XGBoost is a prominent ensemble learning method that, primarily utilizes decision tree structures. It employs gradient boosting, a technique that iteratively refines models by integrating multiple weak learners to formulate a robust predictive framework.

Regularization: A distinctive feature of XGBoost is the incorporation of a regularization term into its objective function. This term is instrumental in mitigating the risk of overfitting, thereby enhancing the model generalization.

Objective Function:

$$\text{Objective}_{XGB}(\theta) = \sum_{i=1}^n \text{loss}(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (17)$$

where θ denotes the model parameters, n is the number of observations, $\text{loss}(y_i, \hat{y}_i)$ is the loss function, and $\sum_{k=1}^K \Omega(f_k)$ is the regularization component.

Random Forest is an ensemble learning technique based on decision tree algorithms. It constructs a multitude of decision trees during training, and their collective output obtained through averaging or majority voting, constitutes the final model prediction.

Overfitting Reduction: The algorithm introduces randomness in tree generation, effectively reducing overfitting compared to individual decision trees.

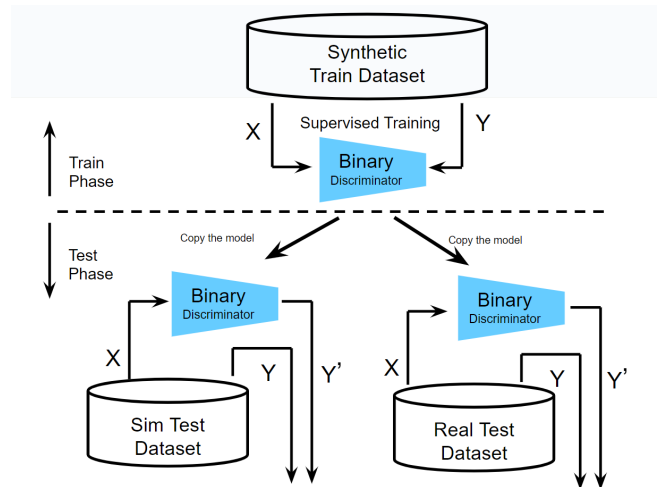


FIGURE 11: Flow of evaluation with Supervised Binary Discriminator

Objective Function:

$$\text{Objective}_{RF}(\theta) = \frac{1}{n} \sum_{i=1}^n \text{loss}(y_i, \hat{y}_i) \quad (18)$$

where θ represents the model parameters, n is the number of data points, and $\text{loss}(y_i, \hat{y}_i)$ are the loss functions.

An **MLP** is a class of feedforward artificial neural networks, characterized by multiple layers of nodes. Each layer is interconnected through weights and biases, enabling MLPs to capture complex, non-linear relationships in the data.

Backpropagation: MLPs rely on backpropagation for training, which is an algorithm that iteratively adjusts weights and biases to minimize the error between the actual and predicted outcomes.

Objective Function:

$$\text{Objective}_{MLP}(\theta) = \frac{1}{n} \sum_{i=1}^n \text{loss}(y_i, \hat{y}_i) + \alpha \sum_{i=1}^{L-1} \|W_i\|^2 \quad (19)$$

where θ denotes the model parameters, n is the number of observations, $\text{loss}(y_i, \hat{y}_i)$ the loss function, α is the regularization parameter, and L the number of network layers.

For each model, parameter estimation can often be described using Maximum Likelihood Estimation (MLE), which for classification problems, involves maximizing the log-likelihood function:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n \log P(y_i | X_i; \theta) \quad (20)$$

where θ represents the parameters, $P(y_i | X_i; \theta)$ is the probability of the target y_i given the input X_i , and $\hat{\theta}$ is the set of parameters that maximizes the likelihood, that is, involves finding the values of hyperparameters, for example, learning rate (η), max-depth or the number of trees ($n_estimators$) that maximize the likelihood of observing the actual data.

We performed a grid search with cross-validation techniques, where the objective function, which is a combination of the loss function and regularization was minimized during training, and selection was made based on the hyperparameter tuning of each model after cross-validation.

We normalized our dataset with a StandardScaler to improve performance and, trained our ETD-based binary discriminator classifier with benign and synthetic attack samples, as shown in Table 6 for Home A. We evaluated the degree to which the classifier is capable of detecting attack instances with real test data. We repeated the same experiment with Homes B and C, on both the simulated and real attack datasets. In the experiment, we used Jupyter Notebook, an open-source web application, written in Python; hence, it was easy to use with TensorFlow. Table 7 lists the parameter values used in the binary classification experiment. We selected the best hyperparameter values by experimenting with grid search. We performed validation through the $fit()$ function using validation data, that is, simulated and real data. After training and testing each home, we calculated the accuracy

and loss based on the number of correctly classified instances.

TABLE 7: Parameters for the binary supervised discriminator

Model	Parameters	Values
XGB	learning rate	0.1
	n_estimators	100
	reg_alpha	0.01
	reg_lambda	1.0
RF	n_estimators	50
	max_depth	10
	min_samples_leaf	1
	min_samples_split	2
MLP	hidden_layer_sizes	50, 100
	activation	tanh
	batch_size	50
	learning_rate_init	0.001

Our framework integrates model selection and feature selection to optimize the machine learning pipeline for ETD. It uses synthetic training data for the initial model training and hyperparameter tuning. The model is then validated on synthetic and real test datasets to ensure that it generalizes well to unseen, real-world data. The framework incorporates MLE for optimizing the objective function, ensuring that the models are well-calibrated, and providing probabilistic outputs that can be interpreted as risk scores for ETD. Table 8 presents the results of our experiments for all homes.

C. ETD BASED ON UNSUPERVISED LEARNING

The process depicted in Figure 12 involves an unsupervised learning model, specifically an autoencoder, which is trained to detect anomalies based on the reconstruction error.

During the training phase, the autoencoder learns parameter θ by minimizing the reconstruction error of the input historical electricity consumption data (benign only) X :

$$\min_{\theta} \|X - \hat{X}(\theta)\|^2 \quad (21)$$

where $\hat{X}(\theta)$ denotes the reconstructed output of the autoencoder, and θ denotes its parameters.

A threshold τ was established based on the reconstruction error distribution during the training phase as shown in Table

TABLE 8: Flow of the evaluation with Synthetic Binary Discriminator

Home	DataSet	Model	Acc	Recall	F1-score	Prec	AUC
A	Sim	XGB	95.21%	0.9169	0.9492	0.9839	98.76%
		RF	91.97%	0.8452	0.9113	0.9886	96.47%
		MLP	97.33%	0.9633	0.9723	0.9816	96.56%
	Real	XGB	95.55%	0.9209	0.9530	0.9875	98.91%
		RF	92.23%	0.8505	0.9146	0.9894	97.02%
		MLP	92.43%	0.9048	0.9212	0.9383	96.68%
B	Sim	XGB	96.15%	0.9281	0.9590	0.9921	98.78%
		RF	93.59%	0.8718	0.9296	0.9957	97.56%
		MLP	95.14%	0.9237	0.9486	0.8718	97.24%
	Real	XGB	96.24%	0.9293	0.9599	0.9927	98.78%
		RF	93.27%	0.8662	0.9259	0.9942	97.64%
		MLP	94.41%	0.9250	0.9413	0.9583	97.95%
C	Sim	XGB	95.26%	0.9116	0.9480	0.9837	98.53%
		RF	92.69%	0.8539	0.9168	0.9906	96.25%
		MLP	91.68%	0.8791	0.9168	0.9406	95.37%
	Real	XGB	95.70%	0.9209	0.9540	0.9894	98.53%
		RF	92.91%	0.8612	0.9216	0.9911	96.54%
		MLP	92.93%	0.9026	0.9251	0.9487	96.49%

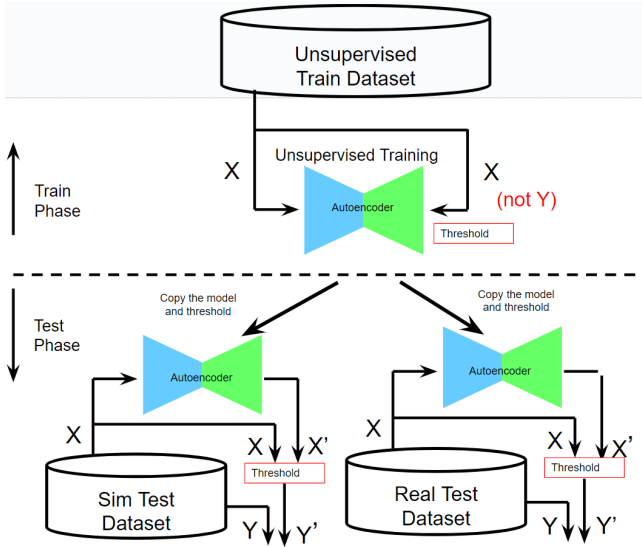


FIGURE 12: Flow of evaluation with Unsupervised Autoencoder

10. These thresholds were used to classify the data points as either normal or anomalous.

In the testing phase, the autoencoder reconstructs new data from both Simulated and Real Test Datasets:-

$$\hat{X}' = \text{Autoencoder}(X') \quad (22)$$

Subsequently, the reconstruction error E for each data point is computed:

$$E = \|X' - \hat{X}'\| \quad (23)$$

An anomaly is flagged if the reconstruction error E exceeds the predetermined threshold τ :

$$Y' = \begin{cases} \text{Anomaly} & \text{if } E > \tau \\ \text{Normal} & \text{if } E \leq \tau \end{cases} \quad (24)$$

The performance of the autoencoder in anomaly detection is contingent on the accuracy of the threshold τ and its capability to accurately learn the representation of normal data during training.

The binary classification output Y' indicates whether a data point is normal or anomalous, based on the reconstruction error relative to the threshold.

Our experimental parameter settings in Table 9 reference attack detection based on unsupervised binary classification models [39]: Multi-Layer Perceptron Autoencoder (MLP_AE) employed a fixed learning rate of 0.001. The autoencoder has a single hidden layer consisting of 32 neurons. A batch size of 32 was used for the training. The model was trained for 100 epochs, and the Adam optimizer was utilized. A validation split of 10% was employed during the training.

1D Convolutional Autoencoder (1D-CONV_AE): A fixed learning rate of 0.001 was used. The latent space dimension was set to 64. A batch size of 32 was used during the training.

The model was trained for 100 iterations. The Adam optimizer was utilized and a validation split of 10% was employed during training. We used 100 trees in the Isolation Forest (IF) algorithm. The random state was set to 42 to ensure reproducibility. The contamination parameter was set to 0.05, which represented the assumed proportion of outliers in the dataset.

TABLE 9: Parameters for legacy unsupervised models

Model	Parameters	Values
MLP_AE	learning rate	0.001
	hidden_layer_sizes	32
	batch_size	32
	epoch	100
	optimizer	adam
	Validation_split	0.1
1D-CONV_AE	learning rate	0.001
	latent_dim	64
	batch_size	32
	epoch	100
	optimizer	adam
	Validation_split	0.1
IF	n_estimators	100
	random state	42
	contamination	0.05

Similar to the supervised binary discriminator, we first normalized the data using MinMaxScaler and trained the autoencoder with historical electricity consumption data X .

Equations (25), (26), (27), and (28) show each that TP, TN, FP, and FN are true positive, true negative, false positive, and false negatives respectively. A TP refers to a sample that is malicious and is detected as malicious. TN indicated a benign sample that was detected as benign. FP indicates that the sample is benign but is detected as malicious. An FN represents a malicious sample that is detected as benign [40].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (25)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (26)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (27)$$

$$F1 - \text{Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (28)$$

For our electricity consumption data from various homes and datasets, normalization facilitates the scaling of input features. This scaling ensures that the features, such as aggregate power consumption, have uniform scales across different homes and datasets; therefore, the original feature values X are transformed into scaled values X_{scaled} within the $[0, 1]$ range. This standardized scaling process is essential for the autoencoder to accurately learn and detect anomalies in electricity consumption patterns.

During training, the autoencoder leverages these scaled features to reconstruct benign data, and a threshold is determined using the statistical method Median Absolute Deviation (MAD), (Equations 29 and 30), on reconstruction errors

to identify anomalies. This feature scaling contributes to the robustness and accuracy of anomaly detection in the context of protecting homeowners from energy theft by identifying unusual electricity consumption patterns, as indicated in the experimental data from different homes in Table 6.

Table 11 shows the results for all the algorithms in different homes. We deployed the receiver operating characteristic curve (AUC-ROC), F1-score, precision, and recall metrics, which are often more informative in such cases. High accuracy can be achieved by simply classifying everything as benign, which does not help in detecting attacks.

In Figure 14 we plotted the ROC curves for the MLP-AE, 1D-CONV-AE, and IF for both simulated and real attack scenarios in Home A. The MLP-AE detector outperformed both 1D-CONV-AE and IF with an AUC score of 0.76 for simulated and 0.59 for real attacks, respectively while 1D-CONV-AE had ROC value 0.67 and 0.61; IF has AUC values 0.64 and 0.54 respectively.

TABLE 10: Legacy unsupervised AE Threshold values for Anomaly detection

Home	Dataset	Model	Threshold
A	Sim & Real	UN-MLP-AE	0.017565
		UN-1D-CONV-AE	0.000020
B		UN-MLP-AE	0.018085
		UN-1D-CONV-AE	0.000055
C		UN-MLP-AE	0.019676
		UN-1D-CONV-AE	0.000055

In the Z-score method, the median is a robust statistic, meaning that it is not significantly affected by outliers. This is called the Robust Z-score method, and instead of using standard deviation, it uses the MAD (Median Absolute Deviation) [41].

The mathematical representation of the Modified Z-score is:

$$\text{MAD} = \text{median}(|x_i - \tilde{x}|) \quad (29)$$

$$\text{Modified Z-score} = 0.6745 \times \frac{x_i - \tilde{x}}{\text{MAD}} \quad (30)$$

0.6745 is the 0.75th quartile of the standard normal distribution, to which the MAD converges.

- $\tilde{x}$ which is just the median of the sample
- MAD, is calculated by taking the absolute difference between each point and the median, and then calculating the median of those differences.

In many real-world scenarios, the attack patterns vary and evolve constantly. Rare attack patterns can be vastly exceeded by benign data. Consequently, the autoencoder may not have sufficient examples of these rare attacks to learn effective representations, making it difficult to detect new attacks.

TABLE 11: Evaluation of the ETD model with AUC and accuracy Scores

Home	DataSet	Model	AUC	ACC
A	Sim	SYN-XGB	98.76%	95.21%
		SYN-RF	96.47%	91.97%
		SYN-MLP	96.56%	97.33%
		UN-MLP-AE	74.61%	58.28%
		UN-1D-CONV-AE	67.31%	54.83%
		UN-IF	63.55%	63.79%
	Real	SYN-XGB	98.91%	95.55%
		SYN-RF	97.02%	92.23%
		SYN-MLP	96.68%	92.43%
		UN-MLP-AE	58.96%	51.26%
		UN-1D-CONV-AE	61.12%	50.51%
		UN-IF	53.66%	53.66%
B	Sim	SYN-XGB	98.78%	96.15%
		SYN-RF	97.56%	93.59%
		SYN-MLP	97.24%	95.14%
		UN-MLP-AE	78.90%	64.58%
		UN-1D-CONV-AE	77.04%	65.63%
		UN-IF	61.64%	62.15%
	Real	SYN-XGB	98.78%	96.24%
		SYN-RF	97.64%	93.27%
		SYN-MLP	97.95%	94.41%
		UN-MLP-AE	60.61%	52.58%
		UN-1D-CONV-AE	71.77%	53.58%
		UN-IF	55.01%	55.01%
C	Sim	SYN-XGB	98.53%	95.26%
		SYN-RF	96.25%	92.69%
		SYN-MLP	95.37%	91.68%
		UN-MLP-AE	75.04%	55.20%
		UN-1D-CONV-AE	64.95%	60.22%
		UN-IF	67.66%	68.82%
	Real	SYN-XGB	98.53%	95.70%
		SYN-RF	96.54%	92.91%
		SYN-MLP	96.49%	92.93%
		UN-MLP-AE	65.64%	50.26%
		UN-1D-CONV-AE	54.95%	51.10%
		UN-IF	74.35%	74.35%

V. ETD EXPERIMENTS

For our investigation, we deployed Python Jupyter Notebook 3.9.16 and Keras [42] with TensorFlow as the back-end. We conducted our experiment using an Intel Core i9 CPU 2.50GHz with 16GB RAM. Windows 11 (64-bit), and NVIDIA GeForce RTX 3070 Ti Laptop GPU.

A. OVERVIEW OF THE FEATURE SET

Each of the appliance datasets has the same features such as Voltage (V), Current(I), Frequency (f), Active power (P), reactive power (Q), apparent power (S), cumulative power over-time (Pt), and power factor (pf). We extracted the three features - P, Q, and, S which play significant roles in understanding electrical appliance patterns and detecting anomalies. The composition of the main primary feature, power consumption, is the combination of these three features. Therefore, the primary feature of our proposed machine learning framework is the aggregated power consumption data, referred to as `power_base`. This dataset is pivotal for our analysis and predictive modeling. The structure of `power_base` is organized as a two-dimensional array, where each en-

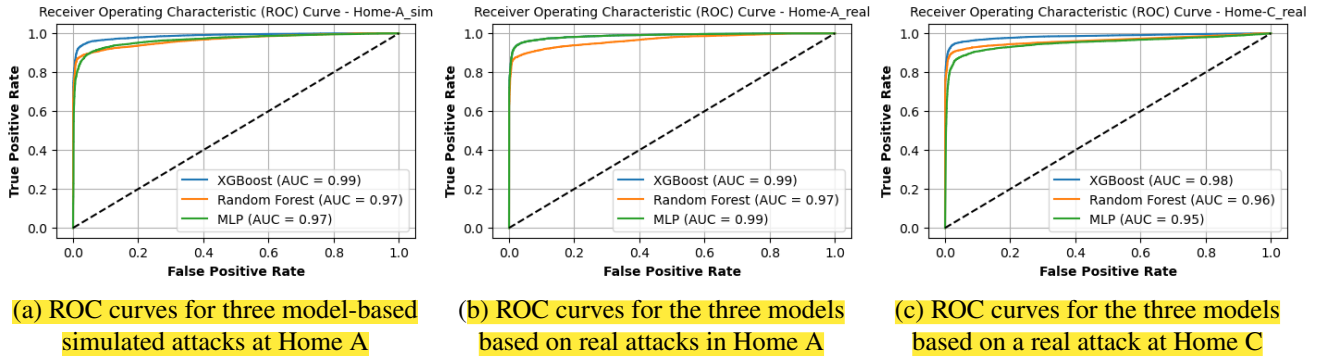


FIGURE 13: Sampled ROC curves comparison for performance evaluation of our proposed synthetic ETD of some selected homes for simulated and real attack.

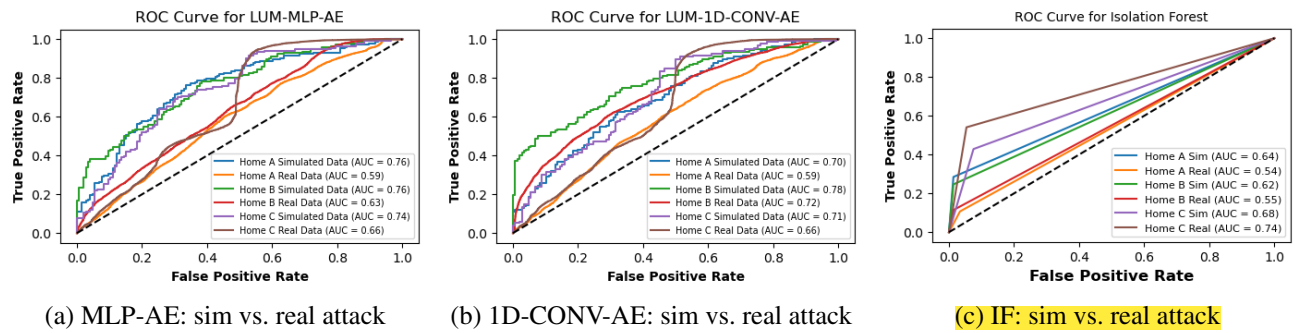


FIGURE 14: Performance comparison of ROC curves for legacy unsupervised models across homes.

TABLE 12: Synthetic Discriminator Parameter for Multi-Class Classification

Model	Parameters	Values
GB	learning rate	0.1
	n_estimators	100
	max_depth	3
XGB	learning rate	0.1
	n_estimators	100
	max_depth	3
	objective	'multi:softprob'
RF	n_estimators	100
	max_depth	None
DTC	random_state	12
MLP-4	hidden_layer_1	720
	hidden_layer_2	200
	activation	'relu'
	solver	'adam'
	learning_rate_init	0.001
MLP-3	hidden_layer_sizes	720
	activation	'relu'
	solver	'adam'
LR	C	1.0
RRC	alpha	1.0
SVM	C	1000
	kernel	'rbf'

try in `power_base[day][min]` denotes the aggregated power consumption for a specific category for a given day and minute. The data spanned a temporal resolution of one minute, summing up to 1440 min (24 h) per day. This granularity allows for a detailed analysis and forecasting of power consumption patterns.

Features description (Independent variables)

1) Aggregated power consumption data:

- **Structure:** The `Power_Base` dataset is a dictionary encompassing aggregated electricity consumption metrics. These metrics were recorded on various days and at different time intervals.
- **Data Representation:** Each element within `power_base[day][min]` encapsulates the total power consumption for a designated category, corresponding to a specific day and minute.

2) Input features (X):

- **Matrix dimensions:** The input feature matrix, denoted by `X`, possesses dimensions of (730, 1440 $\times$ num_categories). Here, '730' represents the total number of days covered, and 'num_categories' signify the different power consumption categories (e.g., len(HPE)).

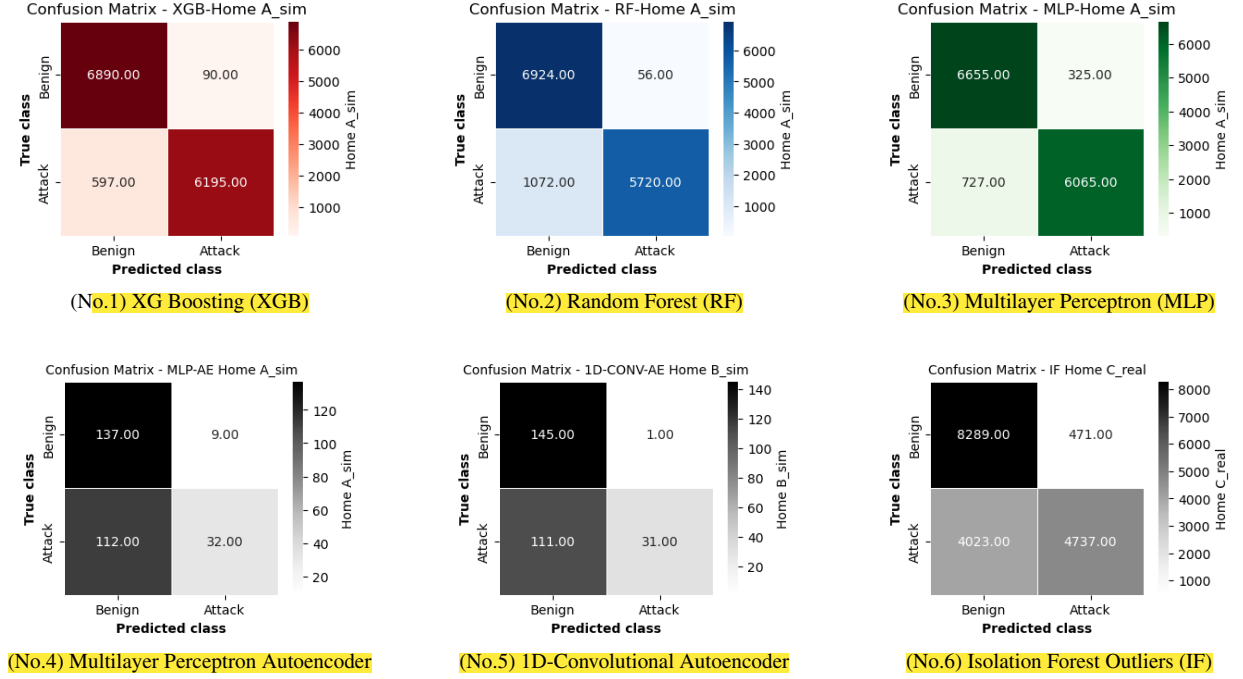


FIGURE 15: Confusion matrices of the trained models. The number (No. #) indicates the rank of the overall accuracy (table 8). The colors indicate the group of accuracy. The attacks class were relatively easily detected by all the synthetic models, especially achieving 99% accuracy with XGB and RF. The false positive rates of XGB, and RF were about 1% which is much better than other models.

- **Data flattening:** The hierarchical structure of `power_base` was transformed into a flattened matrix format. In this matrix, each row symbolizes a day, and each column denotes the aggregated power consumption for a particular category at a specific minute.

3) Output labels (y):

- **Label definition:** The output labels, represented by y , are associated with distinct conditions pertinent to power consumption across the categories specified in the `target_names`.

4) Feature importance assessment:

- **XGBoost feature Scores:** The XGBoost algorithm provides a quantified score reflecting the significance of each feature in the model predictions.
- **Analytical Insights:** By evaluating feature importance, we can ascertain which categories and periods are most influential in predicting conditions related to power consumption.

5) Model Prediction:

- **Application of the Trained Model:** After the training phase, the XGBoost model was employed to forecast power consumption patterns in new, unseen data.

The aggregated power consumption data (`power_base`) served as the primary feature of the proposed (XGBoost) learning framework. The model was trained to predict the

specific conditions related to the power consumption of the target categories (`target_names`). Maximum-likelihood parameter estimation involves optimizing an objective function that combines a loss function and regularization terms as described in Subsection IV, (Equations 17 - 20). The hyperparameters were tuned to optimize the model performance, and feature importance analysis provided insights into the influential factors in the predictions.

VI. EXPERIMENTS RESULTS AND PERFORMANCE EVALUATION

We present a comprehensive performance evaluation of various machine learning models for ETD, utilizing both real-world and synthetic attack datasets. We primarily focus on the AUC metric from Table 11 as the primary evaluation criterion and complement it with additional metrics from Table 8, including the F1-score, accuracy, precision, and recall.

A. MODEL PERFORMANCE BY ROC AND CONFUSION MATRICES

Figures 13 and 14 further generate ROC curves for the remaining supervised benchmark detectors to facilitate a comparative analysis. The ROC curve provides a general view of the model's performance across all thresholds and, provides a sense of discrimination ability. In contrast, the confusion matrix provides detailed information regarding the performance of our model at a specific threshold level.

In the ROC curve, the AUC for each model (XGBoost, Random Forest, MLP) provides a single measure of performance across all possible classification thresholds, summarizing the trade-off between TPR and FPR.

The confusion matrices in Figure 15 provide a more granular view. For instance:

- The MLP for Home A simulated (Home A_sim) confusion matrix indicates that the model correctly identifies 95% of benign cases (TN), the exact value is 6655, and 89% of attack cases (TP), 6065, at a specific threshold.
- The Random Forest confusion matrix showed a high TN rate of 99%, but a lower TP rate of 84%.
- The XGBoost confusion matrix showed a similarly high TN rate of 99% and a better TP rate of 91%.

The ROC curve does not show the actual values of TP, FP, TN, and FN; rather, it shows the rate at which these values change with the different thresholds. A high AUC reflects a model with a high TPR and low FPR across different thresholds, which generally corresponds to high values of TP and TN and low values of FP and FN in the confusion matrices at a particular operating threshold.

The same principles were applied to legacy unsupervised models (LUM). For instance, consider the confusion matrices for the simulated and real data from Home B using the 1D-CONV-AE model in Figure 14(b):

- The AUC of 0.78 and 0.72 for simulated and real data respectively on the ROC curve suggests that the model's ability to distinguish between the classes is reasonably good for simulated data and less so for real data.
- For the corresponding confusion matrices in Figure 15, we see high TN rates (0.99 for simulated, 1.00 for real) but varying TP rates (0.78 for simulated, 0.98 for real). This suggests that, while the model is quite good at identifying negative cases (benign), its performance on positive cases (attacks) is inconsistent between the simulated and real data.
- In the confusion matrices, we observed the specific number of instances that are correctly and incorrectly classified, which was reflected in the ROC curve by the closeness of the curve to the top left corner.

From the ROC curve for the isolated forest model in Figure 14(c):

- Home A Simulated has an AUC of 0.64, meaning that the model has a 64% chance of correctly distinguishing between a benign and an attack instance for the simulated environment of Home A.
- Home A Real had a lower AUC of 0.54, suggesting that the model was less effective in distinguishing between benign and attack instances in the real-world data of Home A.

Similarly, Home B Simulated and Home B Real have AUCs of 0.62 and 0.55, respectively, and Home C Simulated and Home C Real have AUCs of 0.68 and 0.74. The higher the AUC, the better the model is at distinguishing between positive (attacks) and negative (benign) classes. For example,

the model performed best on real data for Home C, with an AUC of 0.74.

From the confusion matrix for Home C real data:

- True Positive (TP): 4737 - The model correctly identified 4737 attack instances.
- True Negatives (TN): 8289 - The model correctly identified 8289 instances as benign.
- False positive (FP): 471 - The model incorrectly identified 471 benign instances as attacks.
- False Negatives (FN): 4023 - The model failed to identify 4023 attacks, mistakenly classified as benign.

Relating the Confusion Matrix to the ROC Curve:

- The specific values in the confusion matrix correspond to a single point on the ROC curve for Home C's real data. The point is determined by the sensitivity (TPR) and FPR.
- These values would indicate a corresponding point on the ROC curve, but the exact point was not marked in the ROC curve. However, we know that point exists, and if the threshold is adjusted, this point moves along the curve, resulting in different values in the confusion matrix.

Generally, the ROC curve indicates how well the model can separate the two classes, and provides a holistic view of the model's performance across all thresholds. By contrast, the confusion matrix indicates exactly where the model makes mistakes at a specific threshold.

B. MODEL TRAINING AND TEST ERROR

To ensure balanced optimization between the training and test errors, we employed GridSearchCV for meticulous hyperparameter tuning. A five-fold cross-validation was incorporated to enhance the robustness of our model evaluations. For the XGBoost model, the regularization parameters L1 (`reg_alpha`) and L2 (`reg_lambda`) were utilized. Conversely, in the context of the Random Forest model, the parameters `min_samples_split` and `min_samples_leaf` serve a regulative function by constraining the complexity of the decision trees. The MLP model employed an L2 penalty term (`alpha`) and a maximum number of iterations (`max_iter`), combined with an early stopping criterion to prevent overfitting.

The subsequent results, as shown in Table 13, were obtained through the post-application of the aforementioned hyperparameter tuning and regularization strategies, as detailed for each model in Table 7.

The bar charts in Figure 16 show the training and test error rates for various models, based on simulated and real data aggregate performance outputs in Table 13. The error rates were calculated as one minus the AUC and accuracy (ACC) values for each model. From the charts, we can observe that for the detection of electricity theft in smart homes utilizing aggregated appliance consumption patterns, the comparative performance analysis of various models is pivotal. Our investigation encompassed the following: both supervised and

TABLE 13: Models aggregate performance with AUC and ACC metric for all homes

Home	DataSet	Model	AUC	ACC
A+B+C	Sim_av	SYN-XGB	98.69%	95.54%
		SYN-RF	96.76%	92.75%
		SYN-MLP	96.32%	94.72%
		UN-MLP-AE	76.18%	59.35%
		UN-1D-CONV-AE	69.77%	60.23%
		UN-IF	64.28%	64.92%
	Real_av	SYN-XGB	98.74%	95.83%
		SYN-RF	97.07%	92.80%
		SYN-MLP	97.04%	93.26%
		UN-MLP-AE	61.74%	51.37%
		UN-1D-CONV-AE	62.61%	51.70%
		UN-IF	64.28%	61.01%

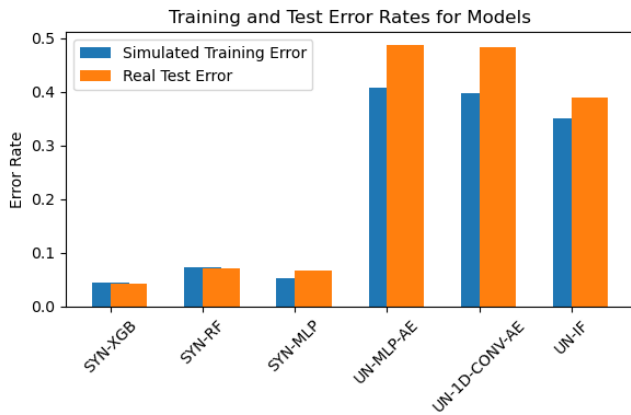


FIGURE 16: Training and test error comparison for our proposed model

unsupervised learning paradigms, with the supervised models demonstrating superior efficacy.

1) Model selection

From our performance analysis with other ETD models for smart homes through aggregated appliance consumption patterns, the XGBoost model (SYN-XGB) emerged as a standout performer, hence it was selected as the best model for our proposed ETD. It achieved the highest Area Under the Curve (AUC) of 98.69% and accuracy (ACC) of 95.54% among the evaluated models. The XGBoost (SYN-XGB) model also exhibited the lowest error rates across simulated and real datasets, indicating its robustness and high accuracy in discerning normal consumption from theft-related anomalies. The robustness of XGBoost is further enhanced by its ensemble learning framework that employs boosting, which has proven effective in managing anomalies and noise prevalent within electricity consumption data. Incorporating regularization within the objective function of the model serves as a bulwark against overfitting, thus facilitating better generalization to unseen data. The model's adeptness in binary classification makes it particularly well-suited for anomaly detection tasks, such as identifying instances of electricity theft in smart home environments. Given its exemplary performance across key

metrics, XGBoost is highly recommended for deployment in ETD systems, where the accurate and reliable identification of theft-related irregularities is paramount.

The Synthetic Random Forest (SYN-RF) and Synthetic Multilayer Perceptron (SYN-MLP) also perform well, although they exhibit marginally higher error rates in comparison to SYN-XGB, suggesting room for optimization.

Conversely, unsupervised models, which include the Unsupervised MLP Autoencoder (UN-MLP-AE), the unsupervised 1D Convolutional Autoencoder (UN-1D-CONV-AE), and the unsupervised IF (UN-IF), exhibit significantly higher error rates. This is particularly notable in scenarios simulating training conditions, which may point to challenges these models face in capturing complex patterns inherent to electricity theft without labeled training data. Notwithstanding, the UN-IF model demonstrates a lesser increase in error rate transitioning from simulated to real datasets, hinting at a certain level of stability in model performance despite lower overall accuracy.

The findings suggest that, in the context of smart home electricity theft detection, supervised models adeptly leverage the nuanced patterns within aggregated appliance consumption data, thus providing a strong foundation for the development of reliable theft detection systems.

Trade-off Between Training and Test Errors

In the dedicated exploration of Electricity Theft Detection (ETD) within smart homes, a critical aspect of our experimental design was ensuring equilibrium between the training and test error rates. This balance is imperative to avert the model's overfitting to training data, which could compromise its generalization capabilities on new, unseen data, a phenomenon that could skew the detection of electricity theft.

Our methodology encompasses the strategic application of GridSearchCV to perform exhaustive hyperparameter tuning, a practice that aids in identifying optimal model parameters. To further bolster the reliability of our findings, we used a five-fold cross-validation scheme, which provides a more rigorous validation of the model's predictive probability.

A regularization schema was adopted for XGBoost algorithm, engaging both L1 (reg_alpha) and L2 (reg_lambda) regularization to mitigate the risk of overfitting. Similarly, with the Random Forest classifier, hyperparameters such as min_samples_split and min_samples_leaf were carefully calibrated to constrain the model complexity and thus, prevent it from learning the noise present in the training data.

The Multilayer Perceptron (MLP) was subjected to L2 regularization (alpha) coupled with a defined maximum number of iterations (max_iter), synergized with an early stopping mechanism. This approach is designed to halt the training process so that the validation performance ceases to improve, thereby circumventing the overfitting problem.

In the domain of unsupervised anomaly detection, particularly when employing autoencoders, the goal is to minimize the reconstruction error across both training and test datasets. The deployment of early stopping techniques, validation data

splits, and potential regularization, although not explicitly utilized in the initial code, form the cornerstone of our strategy to fine-tune the model's complexity. Additionally, the nuanced adjustment of hyperparameters, including the dimensionality of the encoding and hidden layers, as well as the learning rate, was instrumental in achieving a judicious balance between underfitting and overfitting.

A crucial element in our approach is delineating a threshold for anomaly classification based on the reconstruction error obtained from the training data, calibrated with a multiplier of the median absolute deviation. This threshold is pivotal, as it dictates the model's sensitivity and specificity, ultimately influencing the detection accuracy of the ETD system. Such meticulous adjustments are essential for honing the precision of electricity theft detection models tailored to smart home environments.

VII. MODEL PERFORMANCE COMPARISON

A. COMPARISON WITH BENCHMARK MODELS

The comparison of Table 14 with the accuracy and AUC scores and with other classifiers in Table 15 provides a clearer picture of our model performance with benchmark algorithm such as SVM and LR, each model in multiclass and binary classification tasks.

Model	Multiclass Accuracy	Binary Class Accuracy	Multiclass AUC	Binary Class AUC
XGB	0.8450	0.9481	0.9040	0.9675
RF	0.8295	0.9010	0.8762	0.9524
MLP	0.7844	0.8765	0.8520	0.9309
LR	0.7024	0.8041	0.7516	0.8507
SVM	0.7233	0.8211	0.7812	0.8804

TABLE 14: Comparison of models based on accuracy and AUC scores

XGB had the highest scores across the board, achieving 0.85 accuracy and 0.90 AUC in multiclass, and 0.95 accuracy and 0.97 AUC in binary classification. RF is a strong contender with 0.83 multiclass accuracy and 0.88 AUC, (Figure 17(a)), along with 0.90 binary accuracy and 0.95 AUC. Figure 17(b). MLP performs moderately with 0.78 multiclass accuracy and 0.85 AUC, improving binary classification with 0.88 accuracy and 0.93 AUC. LR and SVM, while viable, offer lower accuracy and AUC, suggesting that more advanced methods may be preferable for complex classification challenges.

The line plot presented in Figure 17(c) illustrates the error rates for various evaluation metrics across five different machine learning models: XGBoost (proposed), Random Forest (RF), Multilayer Perceptron (MLP), Support Vector Machine (SVM), and Logistic Regression (LR). The error rate for each model is computed as a $1 - \text{Metric Value}$, where the metrics include accuracy (ACC), Recall, Precision, and Area Under the Curve (AUC). Figure 17(d) shows the aggregate AUC and ACC binary class performance comparison of the proposed models with the benchmark, SVM, and LR models.

TABLE 15: Performance Metrics of Different Multiclass Classification Algorithms

Models	Accuracy	Precision	Recall	F1-score	AUC
GB	0.9342	0.9421	0.9322	0.9334	0.9918
XGB	0.9330	0.9417	0.9291	0.9174	0.9851
RF	0.8652	0.9082	0.8694	0.8701	0.9896
DTC	0.7142	0.7477	0.7068	0.7129	0.8208
MLP-4	0.6610	0.7044	0.6552	0.6812	0.6189
MLP-3	0.6582	0.6877	0.6571	0.6601	0.6097
LR	0.5537	0.5417	0.5543	0.5383	0.5743
RRC	0.5148	0.5280	0.5194	0.5178	0.5060
SVM	0.5220	0.5109	0.5189	0.5085	0.6478

TABLE 16: Aggregate AUC and ACC scores for binary class algorithms for all Homes

Metric	XGB (Proposed)	RF	MLP	SVM	LR
ACC	0.9554	0.9275	0.9331	0.9035	0.8148
Recall	0.9213	0.8581	0.9472	0.8432	0.7311
Precision	0.9882	0.9916	0.9399	0.9471	0.8565
AUC	0.9869	0.9676	0.9632	0.9430	0.8748

- The **XGBoost model (proposed)** shows the most favorable error rates across all metrics, signifying its superior performance relative to the other models.
- The **Random Forest (RF)** and **Multilayer Perceptron (MLP)** models displayed competitive performance, with error rates marginally higher than those of the XGBoost model.
- The **Support Vector Machine (SVM)** and **Logistic Regression (LR)** models exhibit higher error rates, indicating that their performance is not as robust as that of the models above for the tasks evaluated.

The graph underscores the effectiveness of the XGBoost model in minimizing error rates, which correlates with the high predictive accuracy and model reliability for our ETD in smart homes.

VIII. DISCUSSION AND FUTURE WORKS

In this study, we conducted a comprehensive evaluation of machine learning models for electricity theft detection using both real-world and synthetic attack datasets. Our findings shed light on the effectiveness of different models for identifying abnormal consumption patterns indicative of electricity theft. We also consider the implications of these findings for improving the security of home appliance-based electricity consumption monitoring systems.

We contribute to this body of research by systematically evaluating the performance of various ML models, including supervised and unsupervised approaches, using both real and simulated attack data. Our focus on home appliance consumption patterns allows us to gain insight into the strengths and weaknesses of these models in detecting electricity theft scenarios related to specific appliances.

The detection of covert theft schemes that closely mimic legitimate consumption patterns remains challenging. Additionally, scaling these detection systems to accommodate large-scale smart grids and diverse consumer behaviors

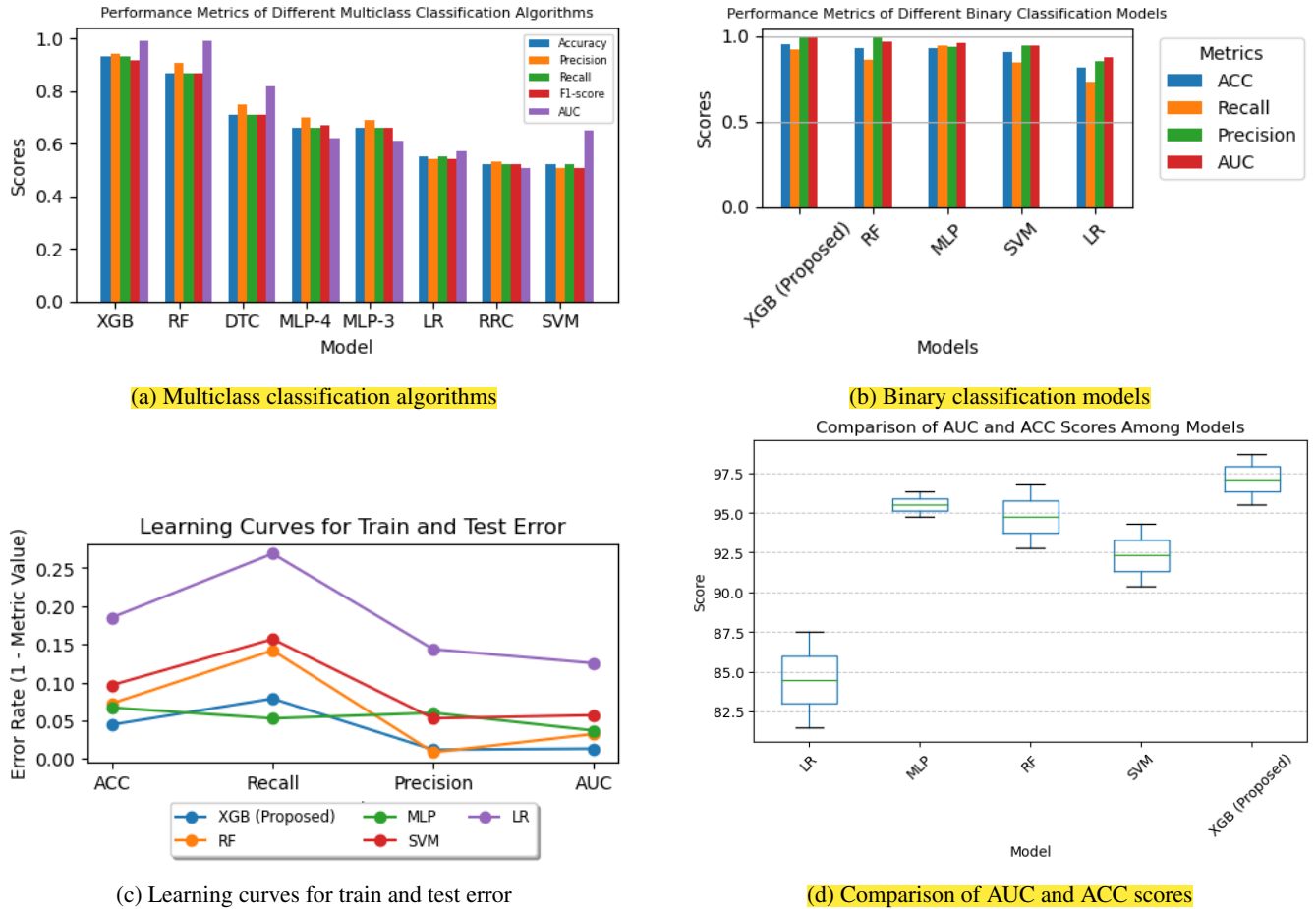


FIGURE 17: Performance metrics of different classification algorithms

presents ongoing challenges.

A. LIMITATIONS OF THE PROPOSED MODELS

Although our models mark significant advancements in detecting electricity theft in smart homes, it is imperative to acknowledge certain limitations that accompany our current methodology.

Dependence on Data Quality and Granularity: The effectiveness of our models is closely tied to the quality and granularity of aggregated appliance consumption data. Any inadequacies in data resolution or representativeness could potentially affect the predictive accuracy of the models.

Differentiation Challenges Between Anomalies and High Usage: One of the inherent challenges our models face distinguishes between legitimate high energy usage scenarios and actual theft instances. This differentiation is crucial to minimize false positives, which could otherwise lead to the misclassification of consumer behavior as fraudulent.

Assumption of Consistent Consumption Patterns: Our models operate under the assumption that consumption patterns within a household remain relatively stable over time. Significant behavioral changes or the introduction of new appliances could alter these patterns, potentially affecting the

model's performance until retraining occurs.

Overfitting Risks and Model Complexity: Despite the implementation of regularization techniques, there remains the risk of overfitting, particularly if the model complexity is not finely calibrated.

Privacy and Ethical Considerations: Although our dataset is anonymized, the utilization of detailed electricity consumption data raises privacy concerns. The potential for re-identification or misuse of this data, even in an anonymized form, cannot be entirely ruled out. Ensuring ongoing compliance with privacy regulations and maintaining ethical standards for data usage is paramount.

Computational Demands and Resource Constraints: The computational complexity associated with our models, especially in terms of hyperparameter tuning and processing, presents limitations in terms of resource allocation.

In future work, we plan to address these limitations knowing well that embarking on further research in these areas will contribute to the ongoing development of robust and effective ETD systems for smart homes. As the field continues to evolve, these challenges provide exciting avenues for future exploration and innovation in the quest for more secure and reliable smart grids.

IX. CONCLUSION

Our study highlights the significance of machine learning models in the detection of electricity theft, particularly concerning home appliance consumption patterns. Supervised learning models, especially the XGB and RF models, exhibited robust performance in various scenarios, making them suitable choices for electricity theft detection tasks. However, these models demonstrate some sensitivity to the complexities of real-world electricity consumption patterns. Unsupervised learning models, although effective in simulated scenarios, struggled to adapt to the intricacies of real attack data, indicating the need for further research to enhance their performance in real-world settings. Overall, our findings provide valuable insights into the application of machine learning in energy security and emphasize the importance of considering home appliance consumption patterns in the detection of electricity theft, benefiting both consumers and energy providers in achieving more efficient and secure energy usage.

CONFLICT OF INTEREST DECLARATION

The authors declare that they have NO affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

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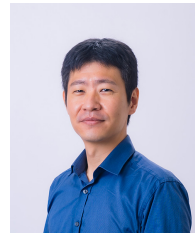


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